

Lumen: An Open, Differentiable, GPU-Parallel Environment for Wall-Safe Endovascular AI

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Abstract

Endovascular agents should be judged on more than whether a device reaches a target branch. They also need wall-safety accounting, reproducible cases, image observations, labels, and episode records that can be inspected after training. Lumen is an Apache-2.0 environment for this work. It provides procedural vascular cases, tube-intrinsic contact, Newton/Warp execution paths, Gymnasium navigation tasks, synthetic fluoroscopy, luminal image generation, and replayable computer-vision datasets. This preprint describes the public release, reports the `lumen-bench/2` launch benchmark, and adds a matched branch-navigation PPO comparison against CathSim. In the 50,000-step PPO comparison, Lumen achieved 100% raw success and 100% safe success across 30 held-out episodes. CathSim achieved 100% raw success but 6.7% safe success under the force threshold used in the comparison. The release is intended to make wall-safe navigation, simulator-derived labels, and reproducible endovascular RL results easier to audit.

1 Motivation

Autonomous catheter and guidewire navigation is now an active research direction for robotic endovascular intervention. Reinforcement learning, imitation learning, inverse reinforcement learning, and expert-trajectory learning have all been tested in simulated or physical settings [4, 5, 1, 2, 3, 6]. CathSim showed that an open simulator could support autonomous catheterization experiments with robot/device simulation and reinforcement-learning tasks [1, 2]. SOFA and BeamAdapter represent another important line of work: finite-element flexible-device modeling for interventional-radiology simulation and mechanical-thrombectomy navigation studies [7, 8, 5].

Lumen is built for the part of the problem that is easy to blur in a demo: scoring the run after it is over. It treats target reach and wall-safe target reach as different outcomes. A policy that reaches a branch after unsafe wall interaction is recorded differently from a policy that reaches the same branch cleanly. The same run can also produce image observations, masks, keypoints, and replay metadata, so the result can be checked outside the RL loop.

2 System Overview

Lumen is organized around five parts:

1. **Procedural vascular cases.** Tube, stenotic, tortuous, and branching vascular cases are generated as replayable scenes, enabling deterministic reruns and benchmark submissions.
2. **Tube-intrinsic contact and wall state.** Device progress is represented along the vessel manifold. Route progress, contact, penetration, torsion, and friction hooks are reported by the environment.

3. **GPU-parallel execution.** Newton/Warp-style kernels support high-throughput differentiable simulation paths for calibration and batched environment work [13].
4. **Observation and dataset generation.** Fluoroscopy, luminal RGB, segmentation masks, keypoints, device masks, and detector-noise variants are generated from the same scene.
5. **Gymnasium-compatible RL tasks.** Navigation tasks are registered through the Gymnasium interface, allowing standard reinforcement-learning tooling to treat Lumen as a conventional environment [14, 15].

This follows the practical pattern used by general RL benchmarks [14, 15], physics-backed control suites [22, 23], and medical-simulation frameworks [7, 9, 10], but the task definitions and scorecards are specific to endovascular navigation.

3 Observation Layer

Lumen emits both state observations and image observations from the same case. A single scene can produce biplanar digitally reconstructed radiographs, masks for vessel and device geometry, labeled keypoints, noisy fluoroscopy, and a luminal RGB camera view. These outputs are meant for reinforcement learning, computer-vision pretraining, perception debugging, and dataset construction.

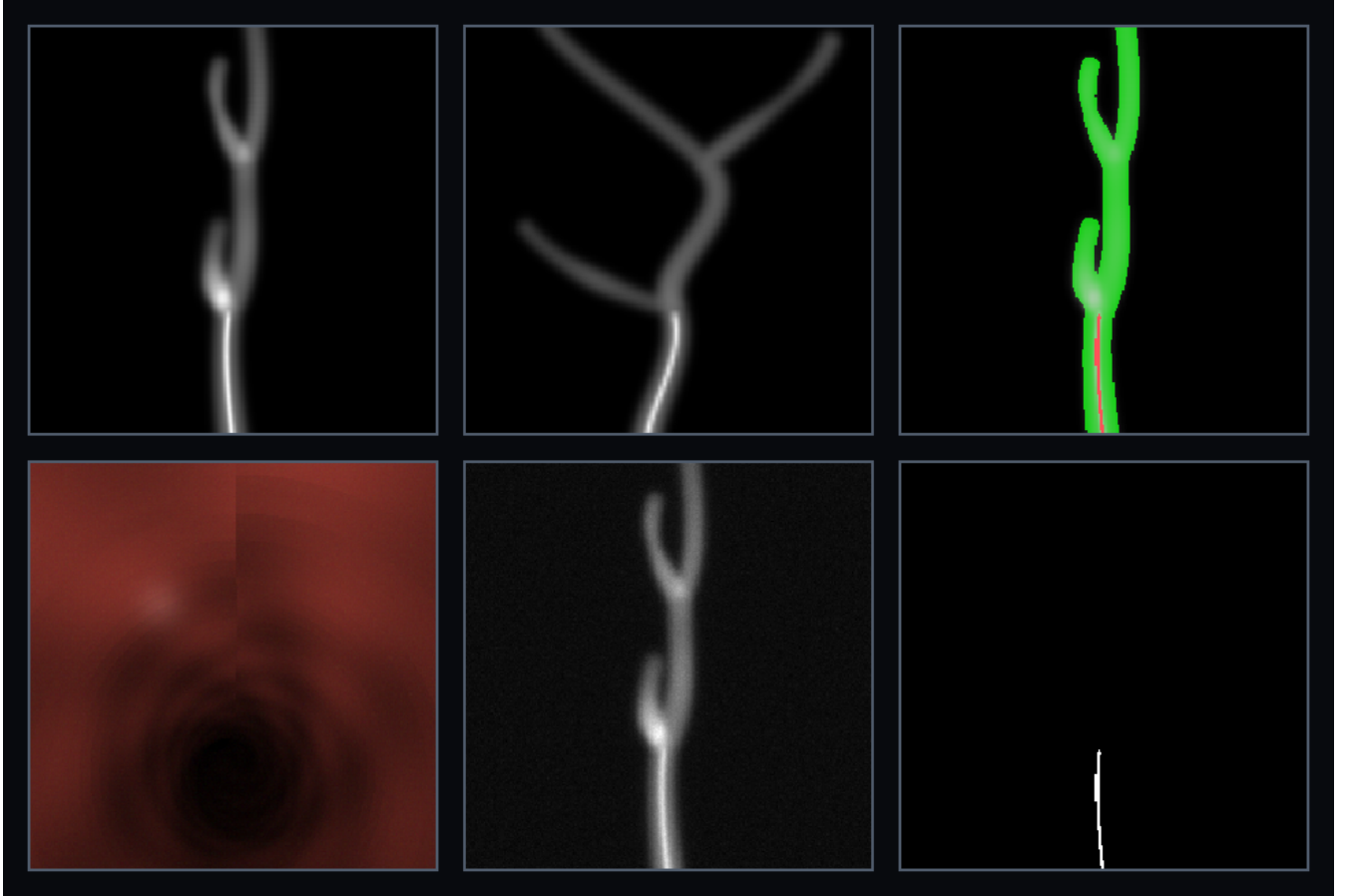


Figure 1: Multi-modal scene outputs from one Lumen case: biplanar fluoroscopy, vessel/device labels, luminal RGB, detector noise, and masks. The same geometry drives control state and image data.

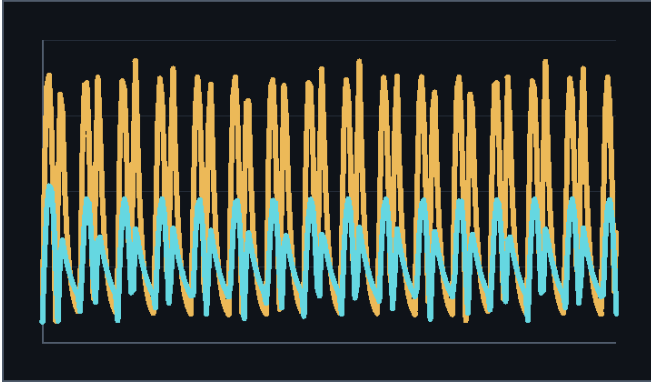
The dataset workflow includes command-line capture, validation, indexing, and train/validation/test

splitting. Episode sidecars record label, mask, and replay metadata before the dataset is consumed by a model.

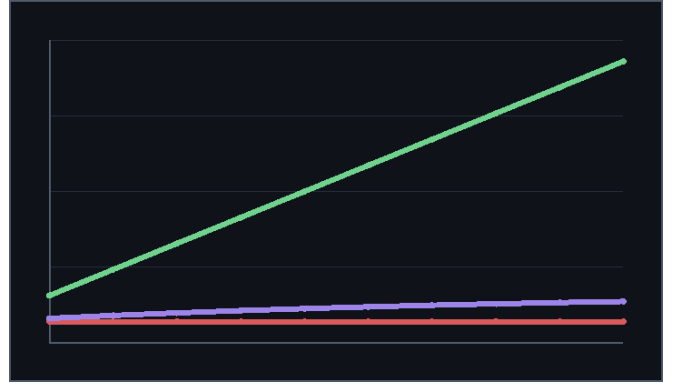
4 Physics and Device Modules

Lumen includes reduced-order modules for events that should affect policy behavior:

- deformable-wall semantics inspired by arterial-wall mechanics and anisotropic constitutive modeling [11, 12];
- contact and penetration metrics reported in environment state;
- tortuosity, route progress, and branch-target scoring;
- flow-diverter effects and aneurysm inflow traces;
- clot-field interaction, stentriever retrieval, fragmentation, and damage bookkeeping;
- synthetic fluoroscopy and luminal rendering for image-observation agents.



(a) Aneurysm inflow before and after a flow-diverter state change.



(b) Stentriever retrieval, captured clot, and damage traces.

Figure 2: Simulator-native advanced module outputs from the current implementation. These are raw Lumen captures, not website screenshots or generated promotional imagery.

These modules keep more of the procedure in the recorded state. Contact, flow, clot/device state, image formation, and replay metadata can be captured together and reproduced from case metadata.

5 Benchmark Protocol and Results

The launch benchmark, `lumen-bench/2`, evaluates three fixed procedural tasks: `nav_tube`, `nav_stenotic`, and `nav_tree_branch`. Each task is run for five seeded episodes. The scorecard ranks safe target reach before raw target reach, then wall safety, then return. The safety threshold for the launch scorecard is a maximum wall penetration of 0.3 simulator units.



Figure 3: Navigation rollouts in branching and tortuous vessels. Lumen reports route progress, target reach, wall penetration, and safety status so benchmark results can distinguish safe target reach from unsafe reach.

Table 1: Launch benchmark result for the deterministic **forward-baseline** policy on **lumen-bench/2**. The scorecard is included with the release artifacts.

Task	Tier	Episodes	Safe success	Unsafe success	Mean steps	Max pen.
nav_tube	Easy	5	1.00	0.00	18.6	0.000
nav_stenotic	Medium	5	1.00	0.00	19.8	0.000
nav_tree_branch	Hard	5	1.00	0.00	54.0	0.178
Overall	–	15	1.00	0.00	–	0.178

The forward baseline is intentionally simple: it advances the proximal end at full rate. It is useful as a reproducibility check and a floor for trained policies, not as a ceiling. Standard reinforcement-learning algorithms such as PPO and SAC can be applied through the Gymnasium interface [16, 17, 18], while leaderboard submissions preserve per-task scorecards for comparison.

To test the Gymnasium training path against the closest open RL comparator, a matched branch-navigation PPO run was executed for Lumen and CathSim. Both environments were trained for 50,000 PPO steps with seed 0 and evaluated for 30 deterministic held-out episodes. Lumen used **nav_tree_branch**. CathSim used **phantom3_bca**. Raw target reach was counted separately from safe target reach. For Lumen, safe reach required target success without breaching the wall-penetration threshold. For CathSim, safe reach required target success without exceeding the 2.0 force threshold used

in the external comparison harness.

Table 2: Matched branch-navigation PPO comparison after 50,000 training steps and 30 deterministic evaluation episodes. Safe success is the primary score because it preserves target reach while rejecting unsafe wall or force events.

Environment	Task	Raw success	Safe success	Unsafe events	Mean steps	Eval steps/s
Lumen	<code>nav_tree_branch</code>	1.00	1.00	0.00	58.0	79.7
CathSim	<code>phantom3_bca</code>	1.00	0.07	0.93	36.8	12.1

The same external-comparison harness also ran 30-episode pilot policies across Lumen, CathSim, and the SOFA/stEVE Docker stack. In these checks, Lumen forward and sweep policies reached 100% raw success on all five native Lumen navigation tasks, with safe success on all forward-policy tasks and four of five sweep-policy tasks. CathSim forward and sweep pilots reached 0% success on the two `phantom3` branch targets. SOFA/stEVE executed BasicWireNav and ArchVariety without crashes; its best pilot rates were 20% success on BasicWireNav and 13.3% success on ArchVariety. The raw per-episode records are included with the release artifacts.

6 Comparison to CathSim and SOFA/BeamAdapter

CathSim, SOFA/BeamAdapter, and Lumen occupy related but distinct positions in the open endovascular-simulation landscape. CathSim is the closest open autonomous-catheterization benchmark reference, with MuJoCo-based robot/device simulation, force sensing, and reinforcement-learning tasks [1, 2]. SOFA/BeamAdapter is a mature flexible-instrument simulation framework for catheters, guidewires, and coils, and has been used in SOFA mechanical-thrombectomy navigation studies [7, 8, 5]. Lumen focuses on a public benchmark path where wall safety, image labels, replayable datasets, and trained-agent scorecards are emitted as ordinary artifacts of a run.

Table 3: Capability positioning of Lumen against commonly cited open endovascular-simulation references.

Surface	Lumen	CathSim	SOFA/BeamAdapter
Primary target	Wall-safe AI benchmark and data generation	Autonomous catheterization simulator and RL tasks	Flexible catheter/guidewire mechanics
RL interface	Gymnasium tasks plus portable scorecards	PPO/SAC-style RL evaluation reported	Used by navigation studies; no common leaderboard
Safety metric	Safe success separated from unsafe reach	Contact/force simulation surface	SOFA-scene contact mechanics
Imaging and labels	Fluoro, luminal RGB, masks, keypoints, noisy detector output	Visual simulation and segmentation surfaces	Scene-dependent rendering and mechanics
Geometry	Replayable tube, stenotic, tortuous, and branch cases	Aorta/robot simulation assets	User-authored SOFA scenes and vascular models
Release emphasis	Reproducible benchmark artifacts, CV-ready captures, Apache-2.0	Open autonomous catheterization simulator	Open mechanics framework and plugin ecosystem

The practical contribution is the packaging: wall-safety scoring, procedural task reproducibility, multi-modal labels, trained-agent evaluation, and replay artifacts are available in one public release.

7 Availability

The public repository is github.com/SeldingerMed/seldinger-lumen. The release includes the benchmark scorecard, simulator-native figures, launch video, source LaTeX, generated PDF, and raw comparison outputs.

8 Conclusion

Lumen provides an open endovascular AI environment where vessel navigation, wall safety, synthetic imaging, labels, and replayable benchmark artifacts are produced from the same run. The launch benchmark establishes a reproducible baseline for safe target reach, and the external comparison records how Lumen, CathSim, and SOFA/stEVE behave under the same pilot and PPO evaluation harnesses.

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